

Motion Pixels — Sandbox Exit Brief

Date: 2026-05-21 **Status:** Sandbox phase complete. Model C frozen. Next phase: real-video dataset expansion.

1. What the sandbox was for

A closed-loop sandbox on a single calibrated recording (MACBA rerun, 2026-05-19) — used to iterate on feature schema, training recipe, and rollout machinery without paying the cost of multi-recording ingest. Everything here runs on `mp-data/processed/rerun_macba_2026-05-19/spatial_v21C/trajectories_encoded.csv`: **52 tracks**, **18,014 rows**, world bounds $\approx 12.9 \times 50.7$ m.

The sandbox is **not** thesis evidence. Its purpose was to answer two questions:

- 1. **Architecture & schema** — what is the right LSTM width + feature set to lock in before we spend ingest effort on more recordings?
- 2. **Are we data-bound or architecture-bound?** — is the per-track held-out performance flat because the model is too weak, or because there isn't enough data?

The exit answer to both is now clear. The remainder of this brief documents it.

2. Two experiments, run back-to-back

Both use the same `TrajectoryLSTM` (hidden 128×2 layers, dropout 0.2, MSE on `(target_du, target_dv)`, window 10, autoregressive 20-step rollout with KDTree-IDW spatial refresh).

2a. `schema_ablation_bridge` — real held-out

52 tracks split into 36 train / 8 val / 9 test. Ports the old `experiments/ablation/` recipe onto the v21C MACBA rerun encoding. Four feature sets:

Model	Features	n
A_motion_only	du, dv, speed, heading_sin/cos, turn_rate	6
B_motion_position	A + u, v	8
C_motion_position_spatial	B + dist_to_obstacle_norm, dist_to_boundary_norm	10
D_full_relational	C + entrance_affinity_norm	11

Model	ADE (m)	FDE (m)	Ang err (rad)	path/GT	ang_ratio	curv_corr
A_motion_only	0.4871	0.8998	0.762	0.424	0.000	0.018
B_motion_position	0.5054	0.9478	1.037	0.583	0.034	-0.048
C_motion_position_spatial	0.5315	0.9809	0.933	0.488	0.016	-0.032
D_full_relational	0.5273	1.0131	1.035	0.723	0.000	-0.078

A/B/C/D are within ~ 0.05 m ADE of each other across 9 held-out tracks. `ang_ratio` ≈ 0 and `curv_corr` ≈ 0 across the board. **Interpretation:** at this sample size, the schema ablation cannot be resolved — all four models collapse to nearly straight rollouts. Held-out ADE is dominated by data scarcity, not feature choice.

2b. `schema_ablation_bridge_overfit10x` — capacity check

Same architecture, same recipe, but every base trajectory is duplicated $\times 10$ (520 tracks, 180,140 rows). Train/val/test share content by construction — this is **memorisation capacity**, not held-out performance. Marked OVERFIT10X on every artefact.

Model	ADE (m)	FDE (m)	Ang err (rad)	path/GT	ang_ratio	curv_corr
A_motion_only	0.1037	0.2594	0.819	0.789	1.072	0.287
B_motion_position	0.0957	0.2491	0.730	0.882	0.989	0.344
C_motion_position_spatial	0.0535	0.1096	0.569	0.852	1.132	0.496
D_full_relational	0.0611	0.1343	0.623	0.862	0.963	0.398

Now the ordering is legible: $A < B < D \approx C$, with C the best memoriser. **Interpretation:** the architecture has the capacity to fit. When the signal-to-noise ratio is high enough, spatial features (`dist_to_obstacle_norm`, `dist_to_boundary_norm`) are the schema increment that buys the biggest jump in fidelity.

3. The headline finding

We are data-bound, not architecture-bound.

The two experiments bracket the gap:

- Same model, same features, same training code.
- On 52 held-out tracks: 0.53 m ADE, near-zero curvature correlation.
- On 520 duplicated tracks: 0.053 m ADE, curvature correlation 0.50, angular ratio ≈ 1 .

The architecture is fine. The schema is fine. The training recipe is fine. The dataset is the lever.

This is **the** finding the sandbox was designed to surface, and it tells us where the next phase must invest.

4. Secondary findings worth keeping

- **D regresses below C in overfit10x.** Adding `entrance_affinity_norm` makes the memoriser **worse** (ADE $0.053 \rightarrow 0.061$, ang_err $0.569 \rightarrow 0.623$). At the current hidden width and dataset, the entrance affinity channel adds noise the model can't gate out. Carry as an open question into the next phase — re-evaluate once dataset volume grows.
 - **Angular error dominates displacement error throughout.** Even the best capacity-check model is at ~ 0.57 rad ($\sim 33^\circ$) mean heading error while ADE is only 5 cm. Position is easy to learn from short windows; heading remains hard. The thesis story should account for this asymmetry, and angular metrics should stay on the dashboard, not get dropped in favour of ADE/FDE alone.
 - **E_full_affordance was correctly skipped.** v21C does not emit `openness_lr_asymmetry`, so the bridge code branched to a graceful skip. Don't reintroduce model E until the encoding actually produces that column.
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5. Exit decision — Model C is frozen

Selected: `C_motion_position_spatial` (10 features: motion kinematics + normalised world position + obstacle/boundary clearance).

Rationale:

- Best capacity-check performance on every metric (ADE, FDE, angular error, curvature correlation, angularity ratio).
- Held-out runs are noise-dominated and do not contradict this choice.
- 10 features keeps inference cheap and the KDTree-IDW rollout simple — `entrance_affinity_norm` adds a third KDTree at every step for unclear benefit.

- The feature set matches the thesis story: ego-motion + where-you-are + what-is-near-you. No relational / global-affordance signals.

Frozen artefacts live in `mp-core/trajectory-prediction/frozen_model_C/`:

- `held_out/` — checkpoint from the bridge experiment (real generalization reference).
- `overfit10x/` — checkpoint from the capacity-check experiment (memorisation reference).
- `README.md` — load instructions, architecture spec, metrics, do-not-edit-in-place note.

Both checkpoints are kept so future readers can see the gap between honest and capacity numbers without having to re-run the sandbox.

6. Open risks carried into the next phase

1. **Calibration parity across recordings.** The v21C encoding bakes in MACBA rerun world bounds and a single top-down .png. New recordings will need their own calibration and reference plan; the normalisation pipeline must not silently collapse different rooms into one coordinate frame.
 2. **Held-out is per-track, not per-recording.** Going forward, the held-out partition should be at *recording* granularity — otherwise we'll re-create the same data-bound situation on a different axis.
 3. **entrance_affinity_norm is on probation.** If it still degrades performance on the expanded dataset, drop it from the schema entirely; don't keep dead features.
 4. **Angular metrics are not in the loss.** The MSE-only objective is what produced the heading-error floor. The next phase should consider an auxiliary angular term (e.g., $1 - \cos(\Delta\theta)$) before claiming the schema has hit its ceiling.
 5. **Sandbox-shaped tests do not catch ingest bugs.** The sandbox uses one CSV. The next phase needs at least a minimal multi-recording smoke test in CI so a broken calibration on recording N doesn't silently propagate.
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7. Next phase — real-video dataset expansion

See `docs/dataset_expansion_plan.md` for the scope/plan doc. In one line: lift the pipeline from "one MACBA rerun" to "a registered set of calibrated recordings" without changing the encoder or the trainer in this phase — just the data layer.

Appendix A — Pointers

- Bridge experiment (held-out):
`mp-core/trajectory-prediction/experiments/schema_ablation_bridge/`
- Overfit10x experiment (capacity):
`mp-core/trajectory-prediction/experiments/schema_ablation_bridge_overfit10x/`
- Frozen Model C: `mp-core/trajectory-prediction/frozen_model_C/`
- Source encoding:
`mp-data/processed/rerun_macba_2026-05-19/spatial_v21C/trajectories_encoded.csv`
- Expansion plan: `docs/dataset_expansion_plan.md`